

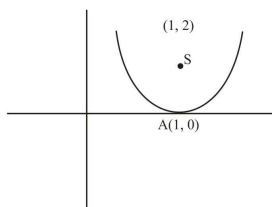
Exercise J01

Q1. The equation of the circle drawn with the focus of the parabola $(x - 1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is

- (A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$ (C) $x^2 + y^2 - 2x - 4y = 0$
 (D) $x^2 + y^2 - 2x - 4y + 1 = 0$

Ans. D

Sol.



$$(x - 1)^2 = 8y$$

$$\text{focus} = \text{center} = (1, 2)$$

$$\text{Radius} = SA = 2$$

$$(x - 1)^2 + (y - 2)^2 = 4$$

Q2. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabola $y^2 = 4ax$ and $x^2 = 4ay$, then

- (A) $d^2 + (3b - 2c)^2 = 0$ (B) $d^2 + (3b + 2c)^2 = 0$ (C) $d^2 + (2b - 3c)^2 = 0$
 (D) $d^2 + (2b + 3c)^2 = 0$

Ans. D

Sol. Given parabolas are

$$y^2 = 4ax \quad \dots (i)$$

$$x^2 = 4ay \quad \dots (ii)$$

Putting the value of y from (ii) in (i), we get

$$\frac{x^2}{16a^2} = 4ax \Rightarrow x(x^3 - 64a^3) = 0 \Rightarrow x = 0, 4a$$

from (ii), $y = 0, 4a$. Let $A \equiv (0, 0)$; $B \equiv (4a, 4a)$

Since, given line $2bx + 3cy + 4d = 0$ passes through A and B,

$$\therefore d = 0 \text{ and } 8ab + 12ac = 0$$

$$\Rightarrow 2b + 3c = 0. (\because a \neq 0)$$

$$\text{Obviously, } d^2 + (2b + 3c)^2 = 0$$

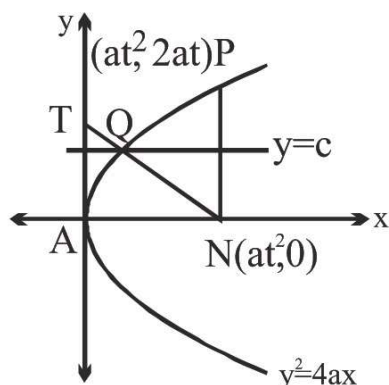
Q3. PN is an ordinate of the parabola $y^2 = 4ax$ (P on $y^2 = 4ax$ and N on x -axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ

meets the tangent at the vertex A in apoint T such that $AT = kNP$, then the value of k is (where A is the vertex)

- (A) $3/2$ (B) $2/3$ (C) 1 (D) $1/3$

Ans. B

Sol.



Equation of PN : $x = at^2$

$y = c$ bisects PN

$$\therefore c = at$$

which cuts the parabola at Q

$$\Rightarrow c^2 = 4ax \quad \therefore x = \frac{c^2}{4a}$$

$$\therefore Q\left(\frac{c^2}{4a}, c\right) = Q\left(\frac{at^2}{4}, at\right)$$

$$\therefore \text{Equation of NQ : } y - 0 = \frac{at - 0}{\frac{at^2}{4} - at^2}(x - at^2)$$

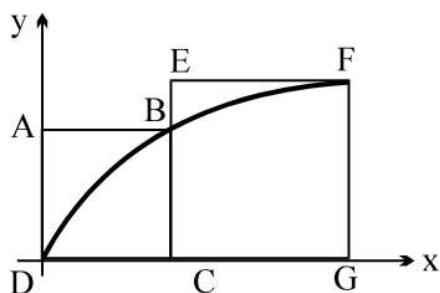
$$y = \frac{-4}{3t}(x - at^2)$$

which cuts $x = 0$ at $\left(0, \frac{4at}{3}\right)$

$$\therefore T = \frac{4at}{3} \text{ and } NP = 2at$$

$$\therefore \frac{AT}{NP} = \frac{\frac{4at}{3}}{2at} = \frac{2}{3} \text{ Ans.}$$

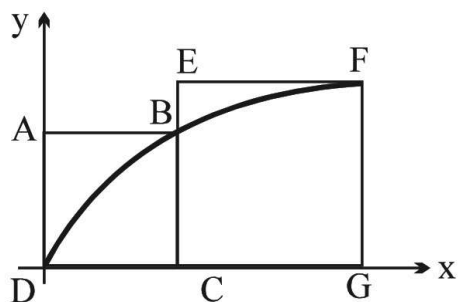
- Q4. ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is



- (A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{3}+1}{4}$

Ans. A

Sol.



$$y^2 = k^2x \Rightarrow y^2 = 4ax \text{ where } k^2 = 4a$$

$$B = (a_1^2, 2at_1); F = (at_2^2, 2at_2) \quad \{t_1 > 0, t_2 > 0\}$$

$$\text{to find } \frac{FG}{BC} = \frac{2at_2}{2at_1} = \frac{t_2}{t_1}$$

$$\text{now } DC = BC \Rightarrow at_1^2 = 2at_1 \Rightarrow t_1 = 2$$

$$\text{also } at_1^2 - at_1^2 = 2at_2$$

$$t_2^2 - 4 = 2at_2$$

$$t^2 - 2t_2 - 4 = 0$$

$$t_2 = \frac{2 \pm \sqrt{4+16}}{2} \text{ but } t_2 > 2 \quad \therefore t_2 \neq \frac{2 - \sqrt{20}}{2}$$

$$t_2 = (\sqrt{5} + 1)$$

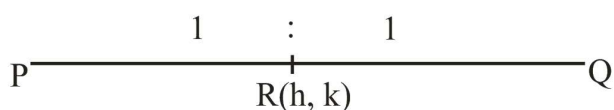
$$\therefore \frac{t_2}{r_1} = \frac{\sqrt{5}+1}{2} \quad \text{Ans.}$$

Q5. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is

- (A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

Ans. A

Sol. $O\left(\frac{t^2}{8}, t\right)$ P(1, 0)



$$h = \frac{\frac{y^2}{8} + 1}{2}$$

$$k = \frac{t}{2}$$

$$(2h - 1)8 = t^2 \quad \dots\dots (1)$$

$$t = 2k \quad \dots\dots (2)$$

from (1) & (2)

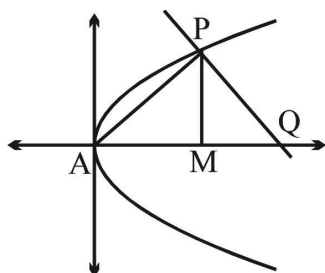
$$(2h - 1)^8 = 4k^2$$

$$y^2 = 2(2x - 1)$$

- Q6. The vertex A of the parabola $y^2 = 4ax$ is joined to any point P on it and PQ is drawn at right angles to AP to meet the axis in Q. Projection of PQ on the axis is equal to
- (A) twice the length of latus rectum (B) the length of latus rectum
(C) half the length of latus rectum (D) one fourth of the length of latus rectum

Ans. B

Sol.



$$A(0, 0), P(at^2, 2at)$$

$$\text{Equation of PQ, } y - 2at = \frac{-t}{2}(x - at^2)$$

$$\Rightarrow \text{coordinates of Q are } (4a + at^2, 0)$$

$$\text{and coordinates of M are } (at^2, 0)$$

$$\Rightarrow \text{length QM} = 4a = \text{Latus rectum} \quad \text{Ans.}$$

- Q7. The graph of the equation $x + y = x^3 + y^3$ is the union of
- (A) line and an ellipse. (B) line and a parabola. (C) line and hyperbola.
(D) line and a point.

Ans. A

$$\text{Sol. Given } (x + y) = (x + y)(x^2 - xy + y^2) \Rightarrow (x + y)(x^2 - xy + y^2 - 1) = 0$$

$$\text{Either } x + y = 0 \text{ or } x^2 - xy + y^2 - 1 = 0$$

Now $x + y = 0$ represent a line.

$$\text{Also } \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 1 & \frac{-1}{2} & 0 \\ \frac{-1}{2} & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = \frac{-3}{4} \neq 0 \text{ and}$$

$$h = \frac{-1}{2}, a = 1 = b, \text{ so } h^2 < ab$$

$$\therefore x^2 - xy + y^2 - 1 = 0 \text{ represent an ellipse.}$$

Hence locus of $P(x, y)$ satisfying above equation is the union of line and an

ellipse. **Ans.**

Q8. A focus of an ellipse is at the origin. The directrix is the line $x = 4$ and the eccentricity is $\frac{1}{2}$. Then the length of the semi-major axis is -

- (A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{8}{3}$

Ans. D

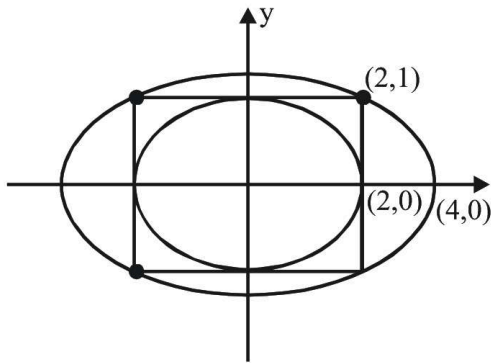
Sol. $e = \frac{5}{2} \Rightarrow \frac{a}{e} - ae = 4 \Rightarrow a\left[2 - \frac{1}{2}\right] = 4 \Rightarrow a \cdot \frac{3}{2} = 4$
 $a = \frac{8}{3} \Rightarrow a\left[\frac{1}{2} - 2\right] = 4$

Q9. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point $(4, 0)$. Then the equation of the ellipse is

- (A) $4x^2 + 48y^2 = 48$ (B) $4x^2 + 64y^2 = 48$ (C) $x^2 + 16y^2 = 16$ (D) $x^2 + 12y^2 = 16$

Ans. D

Sol.



Ellipse $x^2 + 4y^2 = 4$ (Given)

Eqⁿ. of the ellipse (required)

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

Ellipse passes through $(2, 1)$

$$\text{therefore } \frac{4}{16} + \frac{1}{b^2} = 1 \Rightarrow b^2 = \frac{4}{3}$$

$$\frac{x^2}{16} + \frac{y^2}{\frac{4}{3}} = 1 \Rightarrow \frac{x^2}{16} + \frac{3y^2}{4} = 1$$

$$\Rightarrow x^2 + 12y^2 = 16$$

Q10. An ellipse is drawn by taking a diameter of the circle $(x - 1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y - 2)^2 = 4$ as its semi-major axis. If the

centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is :

(A) $x^2 + 4y^2 = 16$ (B) $4x^2 + y^2 = 4$ (C) $x^2 + 4y^2 = 8$ (D) $4x^2 + y^2 = 8$

Ans. A

Sol. Let the equation of ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

from the given conditions

$$a = 4 \text{ and } b = 2$$

$\therefore$ Eq of ellipse is

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$\text{or } x^2 + 4y^2 = 16$$

Q11. For the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$, which of the following remains constant a varies ?

(A) Abscissae of vertices (B) Abscissae of foci (C) Eccentricity (D) Directrix

Ans. B

Sol. The given equation of hyperbola is

$$\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$$

$$\Rightarrow a = \cos \alpha, b = \sin \alpha$$

$$\Rightarrow e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \tan^2 \alpha} = \sec \alpha$$

$$\Rightarrow ae = 1$$

$$\therefore \text{ foci } (\pm 1, 0)$$

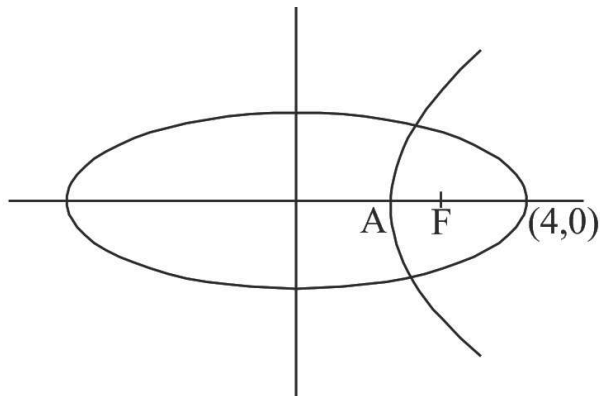
$\therefore$ foci remain constant with respect to α .

Q12. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is.

(A) 5 (B) 7 (C) 9 (D) 4

Ans. B

Sol.



For hyperbola $\frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{81}{25}} = 1$

$$e_H^2 = 1 + \frac{81}{144} = \frac{255}{144}$$

$$e_H = \frac{15}{12} = \frac{5}{4}$$

and $F = ae_H = \frac{12}{5} \cdot \frac{5}{4} = 3$

Hence eccentricity for ellipse $a_E = 3 \Rightarrow a_E = \frac{3}{4}$

$$\therefore \frac{9}{16} = 1 - \frac{b^2}{16} \Rightarrow b^2 = 7 \quad \text{Ans.}$$

Q13. In eccentricity of conjugate hyperbola of the given hyperbola

$$\left| \sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-2)^2} \right| = 3 \text{ is } e', \text{ then value of } 8e' \text{ is}$$

- (A) 12 (B) 14 (C) 17 (D) 10

Ans. D

Sol. For given hyperbola

$$|PA - PB| = 2a$$

$$2a = 3, AB = 2ac \quad A(1, 2) \quad B(5, 8)$$

$$AB = 5$$

$$5 = 3c$$

$$e = \frac{5}{3}$$

eccentricity of conjugate hyperbola : e'

$$\Rightarrow \frac{1}{e^2} + \frac{1}{(e')^2} = 1$$

$$\frac{1}{(e')^2} = 1 - \frac{1}{e^2} = 1 - \frac{9}{25} = \frac{16}{25}$$

$$(e') = \frac{5}{4}$$

$$8e' = 10$$

Q14. A hyperbola, having the transverse axis of length $2 \sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is

(A) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$ (B) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$

(C) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$ (D) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

Ans. A

Sol. Given $3x^2 + 4y^2 = 12$ an ellipse

$$\therefore a^2 = 4 \quad b^2 = 3$$

$$\therefore e = \sqrt{1 - \frac{3}{4}} \Rightarrow e = \frac{1}{2}$$

$\therefore$ It's focus will be $(\pm 1, 0)$

Since hyperbola is confocal to given ellipse, therefore $\pm ae = \pm 1$, but $a = \sin \theta$ given

$$\therefore e = \frac{1}{\sin \theta}, \text{ Now } b^2 = a^2(e^2 - 1)$$

$$b^2 = \sin^2 \theta \frac{\cos^2 \theta}{\sin^2 \theta} \Rightarrow b^2 = \cos^2 \theta$$

Hence required equation will be,

$$\frac{x^2}{\sin^2 \theta} - \frac{y^2}{\cos^2 \theta} = 1$$

$$\Rightarrow x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$$

Q15. Let α and b be non-zero real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents

(A) four straight lines, when $c = 0$ and α, b are of the same sign.

(B) two straight lines and a circle, when $\alpha = b$, and c is of sign opposite to that of a .

(C) two straight lines and a hyperbola, when α and b are of the same sign and c is of sign opposite to that of a

(D) a circle and an ellipse, when α and b are of the same sign and c is of sign opposite to that of α .

Ans. B

Sol. $ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$

either $x^2 - 5xy + 6y^2 = 0 \Rightarrow$ two straight lines passing through origin.

or $ax^2 + by^2 + c = 0$

(A) If $c = 0$, and a and b are of same sign then it will represent a point.

(B) If $a = b$, c is of sign opposite to a then it will represent circle.

(C) When a & b are of same sign and c is of sign opposite to a then it will represent ellipse.

(D) This is clearly incorrect.

Q16. Consider a branch of the hyperbola, $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is

- (A) $1 - \sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}} - 1$ (C) $1 + \sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{2}{3}} + 1$

Ans. B

Sol. The given equation is

$$(x - \sqrt{2})^2 - 2(y + \sqrt{2})^2 = 4$$

$$\frac{(x - \sqrt{2})^2}{4} - \frac{(y + \sqrt{2})^2}{2} = 1$$

$$a = 2, \quad b = \sqrt{2}$$

$$\text{hence eccentricity } e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{3}{2}}$$

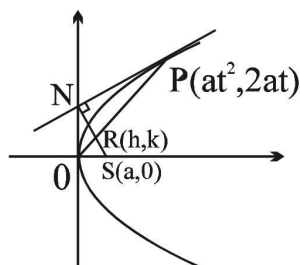
$$\begin{aligned} \text{Area} &= \frac{1}{2}a(e - 1) \cdot \frac{b^2}{a} \\ &= \left(\sqrt{\frac{3}{2}} - 1\right) \text{ sq. units.} \end{aligned}$$

Q17. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is

- (A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$ (C) $2x^2 + 2y^2 - ay = 0$
(D) $2x^2 + y^2 - 2ay = 0$

Ans. B

Sol.



$$T : ty = x + at^2 \quad \dots(1)$$

line perpendicular to (1) through (a,0)

$$tx + y = ta \quad \dots(2)$$

equation of OP : $y - \frac{2}{t}x = 0$ (3)

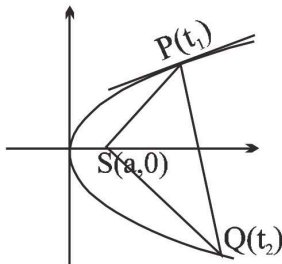
from (2) & (3) eliminating t we get locus]

Q18. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is :

- (A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$

Ans. D

Sol.



$$at_1^2 = 2at_1 \Rightarrow t_1 = 2; P(4a, 4a)$$

$$t_2 = -t_1 - \frac{2}{t_1} = -3$$

$$\therefore Q(9a, -6a)$$

$$\left. \begin{aligned} m_{SP} &= \frac{4a}{4a - a} = \frac{4}{3} \\ m_{SQ} &= \frac{-6a}{9a - a} = -\frac{3}{4} \end{aligned} \right\} \angle PSQ = 90^\circ$$

Q19. PQ is a normal chord of the parabola $y^2 = 4ax$ at P, A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x-axis in R. Then the length of AR is

- (A) equal to the length of the latus rectum
 (B) equal to the focal distance of the point P
 (C) equal to twice the focal distance of the point P
 (D) equal to the distance of the point P from the directrix.

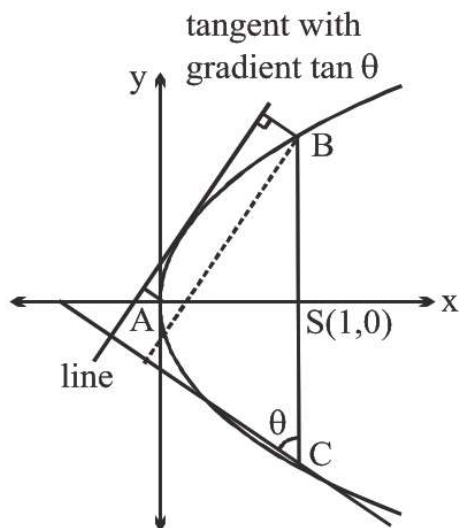
Ans. C

Equation of the line through P parallel to AQ

$$\begin{aligned} \text{put } y = 0 &\Rightarrow x = at_1^2 - at_1t_2 \\ &= at_1^2 - a(-2 - t_1^2) \\ &= 2a + 2at_1^2 = 2(a + at_1^2) \\ &= \text{twice the focal distance of P.} \end{aligned}$$

- (A) 2 (B) $2\sqrt{3}$ (C) $2\sqrt{2}$ (D) $2 + \sqrt{2}$

Sol.



at A, $\theta \rightarrow 90^\circ \Rightarrow \text{projection} = 4$
 at B, $\theta \rightarrow 45^\circ$ or 135°

$$\begin{aligned}\text{Hence } (BC)_{\min.} &= 4 \sin 45^\circ \\ &= \frac{4}{\sqrt{2}} = 2\sqrt{2} \quad \text{Ans.}\end{aligned}$$

Q21. The normal at the point $(bt_1^2, 2bt_1)$ on a parabola meets the parabola again in the point $(bt_2^2, 2bt_2)$, then

(A) $t_2 = t_1 + \frac{2}{t_1}$ (B) $t_2 = -t_1 - \frac{2}{t_1}$ (C) $t_2 = -t_1 + \frac{2}{t_1}$ (D) $t_2 = t_1 - \frac{2}{t_1}$

Ans. B

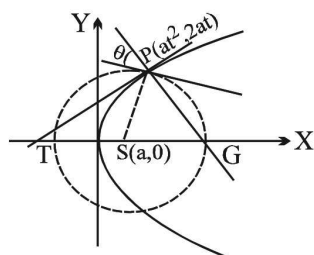
Sol. It is a fundamental theorem.

Q22. The tangent and normal at $P(t)$, for all real positive t , to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at p to the parabola is inclined to the tangent at p to the circle passing through the points P, t and G is

(A) $\cot^{-1}t$ (B) $\cot^{-1}t^2$ (C) $\tan^{-1}t$ (D) $\tan^{-1}t^2$

Ans. C

Sol.



$$\text{slope of tangent} = \frac{1}{t}(m_1) \text{ at } P \text{ on parabola}$$

$$\text{slope of PS} = \frac{2at}{a(t^2 - 1)} = \frac{2t}{t^2 - 1} \text{ (normal to the circle)}$$

$$\therefore \text{slope of tangent at } P \text{ on circle} = \frac{1 - t^2}{2t}(m_2)$$

$$\therefore \tan \theta = \frac{\frac{1}{t} - \frac{1-t^2}{2t}}{1 + \frac{1-t^2}{2t^2}} = t$$

$$\therefore \theta = \tan^{-1}t \Rightarrow (C)$$

Q23. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at $(4, -4)$ and $(9a, -6a)$ is

(A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$ (C) $5x - y - 115 = 0$

(D) $5x + y + 115 = 0$

Ans. B

Sol. Equation of normal at $(4, -4)$ is $y = 2x - 12$ and equation of normal at $(9, -6)$ is $y = 3x - 33$. Point of intersection of both these normals is $(21, 30)$

$$\text{Now } y = mx - 2m - m^3$$

$$30 = 21m - 2m - m^3 = 19m - m^3$$

$$\Rightarrow m^3 - 19m + 30 = 0$$

$$m = 2, 3, -5.$$

Q24. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is :

$$(A) x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0 \quad (B) x^2 + y^2 - 4x + 9y + 18 = 0$$

$$(C) x^2 + y^2 - 4x + 8y + 12 = 0 \quad (D) x^2 + y^2 - x + 4y - 12 = 0$$

Ans. C

Sol. $P(2t^2, 4t)$

Normal at P passes through $(0, -6)$

$$tx + y = 4t + 2t^3$$

$$-6 = 4t + 2t^3$$

$$t^3 + 2t + 3 = 0$$

$$t = -1$$

$$P(2, -4)$$

Q25. If the tangent at $(1, 7)$ to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ then the value of c is

$$(A) 85 \quad (B) 95 \quad (C) 195 \quad (D) 185$$

Ans. B

$$\text{Sol. } x^2 = y - 6$$

$$\left(\frac{dy}{dx}\right)_{(1,7)} = 2$$

$$T(1, 7) = 2x - y + 5 = 0$$

distance of center from tangent = radius

$$r = \left| \frac{-16 + 6 + 5}{\sqrt{5}} \right| = \sqrt{5}$$

$$\sqrt{g^2 + f^2 - c} = \sqrt{5}$$

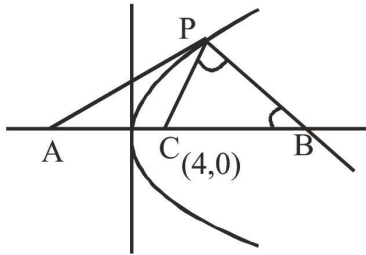
$$\Rightarrow c = 95$$

Q26. Tangent and normal are drawn at $P(16, 16)$ on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If C is the centre of the circle through the points P, A and B and $\angle CPB = \theta$, then a value of $\tan \theta$ is

- (A) 3 (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) 2

Ans. D

Sol.



$$\left(-\frac{dx}{dy}\right) = -t = -1$$

Slope of normal at P = $-\tan\theta = -2$

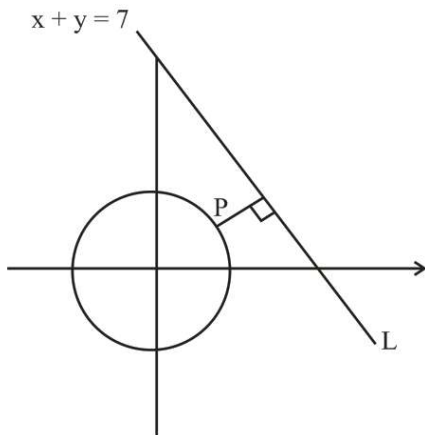
$$\tan\theta = 2$$

Q27. The point on the ellipse $x^2 + 2y^2 = 6$, whose distance from the line $x + y = 7$ is minimum is

- (A) (2, 3) (B) (2, 1) (C) (1, 0) (D) None of these

Ans. B

Sol.



$$x^2 + 2y^2 = 6$$

$$\frac{x^2}{6} + \frac{y^2}{3} = 1$$

$$P(\sqrt{6} \cos \theta, \sqrt{3} \sin \theta)$$

shortes distance will be along common normal

Hence m tanentat p = m line

$$-\sqrt{\frac{3}{6}} \cot \theta = 1$$

$$\cot \theta = \sqrt{2}$$

$$\sin \theta = \frac{1}{\sqrt{3}}, \quad \cos \theta = \frac{\sqrt{2}}{\sqrt{3}}$$

$$P\left(\sqrt{6} \cdot \frac{\sqrt{2}}{\sqrt{3}}, \sqrt{3} \cdot \frac{1}{\sqrt{3}}\right)$$

$$P(2, 1)$$

Q28. The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is :

- (A) $(x^2 + y^2)^2 = 6x^2 + 2y^2$ (B) $(x^2 + y^2)^2 = 6x^2 - 2y^2$ (C) $(x^2 - y^2)^2 = 6x^2 + 2y^2$
 (D) $(x^2 - y^2)^2 = 6x^2 - 2y^2$

Ans. A

Sol. $k = mh \pm \sqrt{6m^2 + 2} \quad \dots\dots(1)$

$$\left(\frac{k}{h}\right)m = -1 \quad \dots\dots(2)$$

Eliminating m from (1) and (2)

Locus is

$$(x^2 + y^2)^2 = 6x^2 + 2y^2$$

Q29. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is-

- (A) an ellipse (B) a circle (C) a parabola (D) a hyperbola

Ans. D

Sol. On comparing $y = mx + \sqrt{a^2m^2 - b^2}$ and $y = \alpha x + \beta$

$$\therefore \alpha = m, \beta = \sqrt{a^2m^2 - b^2}$$

$$\Rightarrow \beta^2 = a^2\alpha^2 - b^2$$

$$\Rightarrow a^2\alpha^2 - \beta^2 = b^2$$

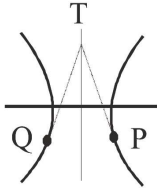
$\Rightarrow$ Hyperbola.

Q30. Tangent are drawn to the hyperbola $4x^2 - y^2 = 36$ at the points P and Q. If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is

- (A) $60\sqrt{3}$ (B) $36\sqrt{5}$ (C) $45\sqrt{5}$ (D) $54\sqrt{3}$

Ans. C

Sol.



Equation of tangent at P

$$\frac{x \sec \theta}{3} - \frac{y \tan \theta}{6} = 1$$

it will pass through (0, 3)

$$0 - \frac{3 \tan \theta}{6} = 1$$

$$\tan \theta = -2 \Rightarrow \sec \theta = \sqrt{5}$$

$$\begin{aligned} \text{area} &= \frac{1}{2} |6 \sec \theta| (|6 \tan \theta| + 3) \\ &= 45\sqrt{5} \end{aligned}$$

- Q31. Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x-axis at (9, 0), then the eccentricity of the hyperbola is

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

Ans. B

Sol. Equation of normal at P(6, 3) on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{a^2 x}{6} + \frac{b^2 y}{3} = a^2 e^2$$

It intersects x-axis at (9, 0)

$$\Rightarrow a^2 \frac{9}{6} = a^2 e^2 \Rightarrow e = \sqrt{\frac{3}{2}}$$

- Q32. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to

(A) $\frac{3\sqrt{2}}{4}$ units (B) $\frac{5\sqrt{2}}{4}$ units (C) $\frac{7\sqrt{2}}{4}$ units (D) $\frac{\sqrt{2}}{4}$ units

Ans. A

Sol. General point on the curve $y^2 = x - 1$ is $(t_1^2 + 1, t_1)$ and the general point on the curve $x^2 = y - 1$ is $(t_2, t_2^2 + 1)$.

Since both the curves are symmetrical about the line $y = x$, for the nearest point on the curve $y^2 = x - 1$ from the line $y = x$

Let, $Q(t_1^2 + 1, t_1)$

$$P(t_2, t_2^2 + 1)$$

slopes of the tangents are same

$$\frac{1}{2t_1} = 2.t_2 t_1 t_2 = \frac{1}{4} \dots\dots (1)$$

Since PQ is perpendicular to line $x = y$

$$\frac{t_1 - t_2^2 - 1}{t_1^2 + 1 - t_2} = -1 \dots\dots (2)$$

Solving (1) & (2), we get $t_1 = t_2 = \frac{1}{2}$

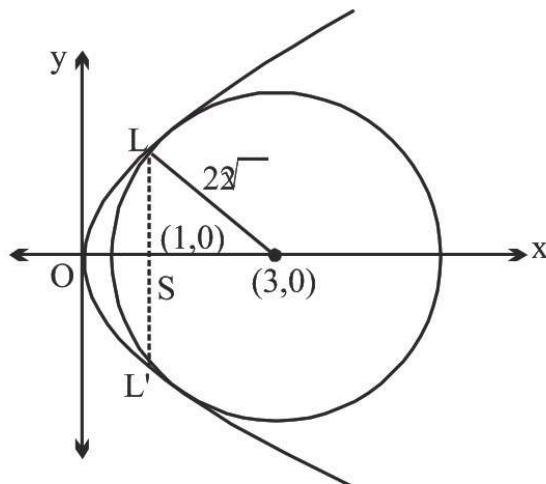
$$P\left(\frac{1}{2}, \frac{5}{4}\right), Q\left(\frac{5}{4}, \frac{1}{2}\right) \text{ therefore, } PQ = \frac{3\sqrt{2}}{4}$$

Q33. Consider two curves $C_1: (y - \sqrt{3})^2 = 4(x - \sqrt{2})$ and $C_2: x^2 + y^2 = (6 + 2\sqrt{2})x + 2\sqrt{3}y - 6(1 + \sqrt{2})$, then

- (A) C_1 and C_2 touch each other only at one point.
- (B) C_1 and C_2 touch each other exactly at two points.
- (C) C_1 and C_2 intersect (but do not touch) at exactly two points.
- (D) C_1 and C_2 neither intersect nor touch each other.

Ans. B

Sol.



$$\text{Given } C_1 : (y - \sqrt{3})^2 = 4(x - \sqrt{2}) \dots\dots(1)$$

$$\text{and } C_2 : (x - (3 + \sqrt{2}))^2 + (y - \sqrt{3})^2 = (3 + \sqrt{2})^2 + 3 - 6(1 + \sqrt{2})$$

$$\left[(x - \sqrt{2}) - 3 \right]^2 + (y - \sqrt{3})^2 = 8 \dots\dots(2)$$

$$\text{Let } y - \sqrt{3} = Y \text{ and } X - \sqrt{2} = X$$

$$C_1' = Y^2 = 4X \dots\dots(3)$$

$$C_2' = (X - 3)^2 + Y^2 = 8 \dots\dots(4)$$

$$\text{Solving } (X - 3)^2 + 4X = 8$$

$$X^2 - 2X + 1 = 0$$

$$(X - 1)^2 = 0$$

$$X = 1$$

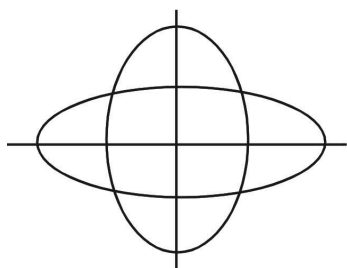
Q34. Equation of the common tangent to the ellipses, $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{a^2 + b^2} = 1$, is

(A) $ay = bx + \sqrt{a^4 - a^2b^2 + b^4}$ (B) $by = ax - \sqrt{a^4 + a^2b^2 + b^4}$

(C) $by = bx - \sqrt{a^4 + a^2b^2 + b^4}$ (D) $by = ax + \sqrt{a^4 - a^2b^2 + b^4}$

Ans. B

Sol.



Equation of a tangent to $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$

$$y = mx \pm \sqrt{(a^2 + b^2)m^2 + b^2} \quad \dots(1)$$

If (1) is also a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{a^2 + b^2} = 1$ then

$$(a^2 + b^2)m^2 + b^2 = a^2m^2 + a^2 + b^2 \quad (\text{using } c^2 = a^2m^2 + b^2)$$

$$b^2m^2 = a^2 \Rightarrow m^2 = \frac{a^2}{b^2} \Rightarrow m = \pm \frac{a}{b}$$

$$y = \pm \frac{a}{b}x \pm \sqrt{(a^2 + b^2)\frac{a^2}{b^2} + b^2}$$

$$by = \pm ax \pm \sqrt{a^4 + a^2b^2 + b^4}$$

Note : Although there can be four common tangents but only one of these appears in B.

Q35. Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are

(A) $x = \pm (y + 2a)$ (B) $y = \pm (x + 2a)$ (C) $x = \pm (y + a)$ (D) $y = \pm (x + a)$

Ans. B

Sol. Let slope of common tangent is m

$$y = mx \pm \sqrt{2a^2m^2 + 2a^2}, \quad y = mx + \frac{2a}{m}$$

$$\sqrt{2a^2m^2 + 2a^2} = \frac{2a}{m}$$

$$2a^2n^2 + 2a^2 = \frac{4a^2}{n^2}$$

$$m^4 + m^2 - 2 = 0$$

$$m^2 = -2, \quad m^2 = 1$$

$$m = \pm 1$$

$$y = \pm(x + 2a)$$

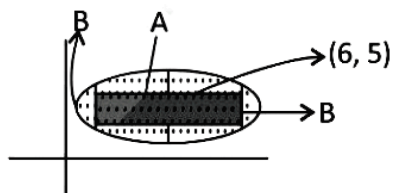
Q36. Two sets A and B are as under :

$A = \{(a, b) \in \mathbb{R} \times \mathbb{R} : |a - 5| < 1 \text{ and } |b - 5| < 1\}$ and $B = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 4(a - 6)^2 + 9(b - 5)^2 \leq 36\}$. Then

(A) $A \cap B = \phi$ (an empty set) (B) neither $A \subset B$ nor $B \subset A$ (C) $B \subset A$ (D) $A \subset B$

Ans. D

Sol. Let $a - 6 = x$ $b - 5 = y$
 $|x + 1| < 1$ $|y| < 1$ $4x^2 + 9y^2 \leq 36$
 $-2 < x < 0$ $-1 < y < 1$



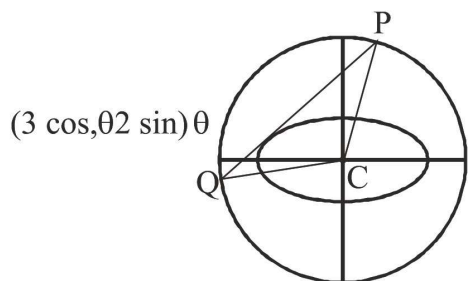
Boundary of square is lying in ellipse
 $\Rightarrow A \subset B$

Q37. A tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is

(A) $\frac{9}{4}$ (B) $-\frac{4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$

Ans. B

Sol.



The equation of the tangent at $(3 \cos \theta, 2 \sin \theta)$ on $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is

$$\frac{x}{3} \cos \theta + \frac{y}{2} \sin \theta = 1 \quad \dots (i)$$

The equation of the director circle is

$$x^2 + y^2 = 9 + 4 = 13 \quad \dots (ii)$$

The combined equation of CP and CQ is obtained by homogenising equation (ii) with (i). Thus combined equation is

$$x^2 + y^2 = 13 \left(\frac{x}{3} \cos \theta + \frac{y}{2} \sin \theta \right)^2$$

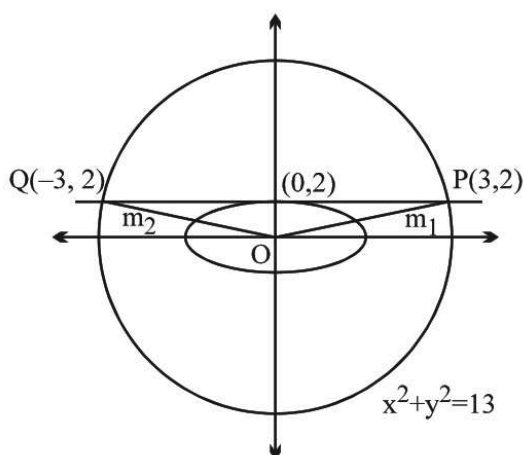
$$\Rightarrow \left(\frac{13}{9} \cos^2 \theta - 1 \right) x^2 + \frac{13}{3} \sin \theta \left(\frac{13}{4} \sin^2 \theta - 1 \right) y^2 = 0$$

$\therefore$ Product of the slopes of CP and CQ

$$\frac{\text{coefficient of } x^2}{\text{coefficient of } y^2} = \frac{\frac{13}{9} \cos^2 \theta - 1}{\frac{13}{4} \sin^2 \theta - 1} = \frac{13 \cos^2 \theta - 9}{13 \sin^2 \theta - 4} \times \frac{4}{9} = \frac{13 \cos^2 \theta - 9}{13 - 13 \cos^2 \theta - 4} \times \frac{4}{9} = -\frac{4}{9}$$

$$[\text{Note: } m_1 + m_2 = \frac{-2h}{b} ; m_1 m_2 = \frac{a}{b}]$$

Alternatively: For objective take tangent on ellipse as $y = 2$



Clearly $m_1 m_2 = \frac{-4}{9}$ **Ans.**

Q38. The normal at a variable point P on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that :

(A) e' is independent of e (B) $e' = 1$ (C) $e' = e$ (D) $e' = 1/e$

Ans. C

Sol. $P(a \cos \theta, b \sin \theta)$

Normal at 'P'

$$\frac{a^2 x}{x} - \frac{b^2 y}{y_1} = a^2 - b^2$$

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = (ae)^2$$

$$Q\left(\frac{(ae)^2 \cos \theta}{a}, 0\right) R\left(0, -\frac{(ae)^2 \sin \theta}{b}\right)$$

Let mid point of QR is 'S' (h, k)

$$h = \frac{(ae)^2 \cos \theta}{2a}$$

$$\frac{2ah}{a^2 c^2} \cos^2 \theta \quad \dots (1)$$

$$k = \frac{-(ae)^2 \sin \theta}{2b}$$

$$\frac{2bk}{a^2 c^2} = -\sin \theta \quad \dots (2)$$

$$(1)2 + (2)2$$

$$\frac{4h^2}{a^2 e^4} + \frac{b^2 4k^2}{a^4 e^4} = 1 \Rightarrow \frac{x^2}{1} + \frac{y^2}{\left(\frac{a}{b}\right)^2} = \frac{e^4 a^2}{4}$$

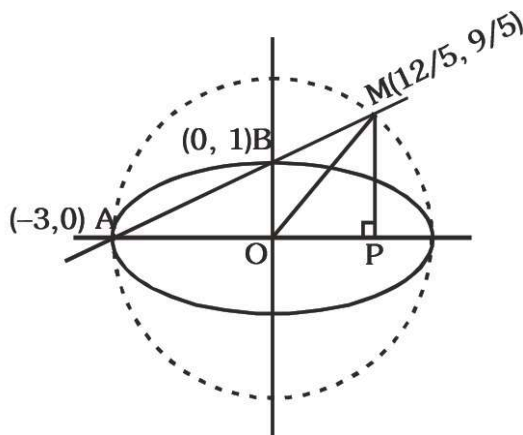
$$e^4 = \sqrt{1 - \frac{1}{\left(\frac{a}{b}\right)^2}} = \sqrt{1 - b^2} = e$$

Q39. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is

(A) $\frac{31}{10}$ (B) $\frac{29}{10}$ (C) $\frac{21}{10}$ (D) $\frac{27}{10}$

Ans. D

Sol.



$$\text{Area of triangle AOM} = \frac{1}{2} \cdot 3 \cdot \frac{9}{5} = \frac{27}{10}$$

$$\Rightarrow \text{Equation of AM is } y = \frac{1}{3}(x + 3)$$

$x - 3y + 3 = 0$ which is chord of auxiliary circle

$x^2 + y^2 = 9$, and PM is ordinate of point M

$$\Rightarrow (3y - 3)^2 + y^2 = 9 \Rightarrow y = \frac{9}{5} = \text{PM} \Rightarrow \text{Area of triangle} = \frac{1}{2} \cdot 3 \cdot \frac{9}{5} = \frac{27}{10}$$

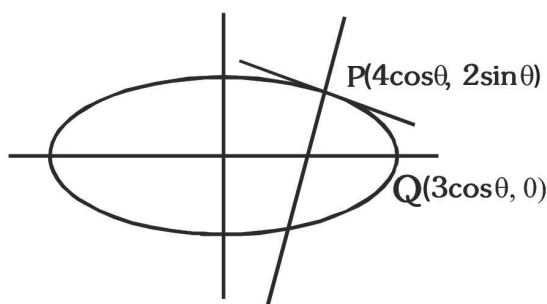
Q40. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the point

(A) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{2}{7} \right)$ (B) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{\sqrt{19}}{4} \right)$ (C) $\left(\pm 2\sqrt{3}, \pm \frac{1}{7} \right)$

(D) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7} \right)$

Ans. C

Sol.



$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$4x \sec \theta - 2y \operatorname{cosec} \theta = 12$$

$$x = 3 \cos \theta$$

$$Q \equiv (3 \cos \theta, 0)$$

$$2h = 7 \cos \theta$$

$$2k = 2 \sin \theta$$

$$\frac{4x^2}{49} + \frac{y^2}{1} = 1 \quad \dots (i)$$

$$\text{L.R} \Rightarrow x = 2\sqrt{3}$$

Putting in (i)

$$y = \frac{1}{7} \quad \left(\pm 2\sqrt{3}, \pm \frac{1}{7} \right)$$

Q41. If α & β are the eccentric angles of the extremities of a focal chord of an standard ellipse, then the eccentricity of the ellipse is :

(A) $\frac{\cos \alpha + \cos \beta}{\cos(\alpha + \beta)}$ (B) $\frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$ (C) $\frac{\cos \alpha - \cos \beta}{\cos(\alpha - \beta)}$ (D) $\frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$

Ans. D

Sol. $\frac{x}{a} \cos \frac{\alpha - \beta}{2} + \frac{y}{b} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}; \frac{ae}{a} \cos \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}$

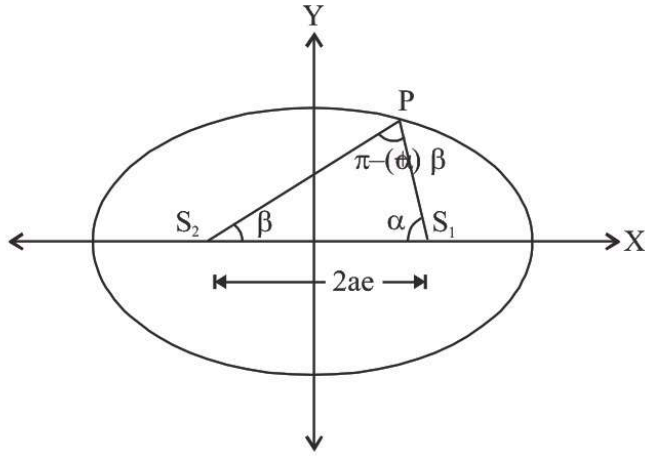
$$\Rightarrow e = \frac{\cos \frac{\alpha - \beta}{2}}{\cos \frac{\alpha + \beta}{2}} \cdot \frac{2 \sin \frac{\alpha + \beta}{2}}{2 \sin \frac{\alpha + \beta}{2}} = \frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$$

Q42. If P is any point on ellipse with foci S_1 & S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha$, $\angle S_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, then $\cot \frac{\alpha}{2}$, $\cot \frac{\gamma}{2}$, $\cot \frac{\beta}{2}$, are in

(A) A.P. (B) G.P. (C) H.P. (D) NOT A.P., G.P. & H.P.

Ans. A

Sol.



By sine rule in $\triangle PS_1S_2$, we get $\frac{2ae}{\sin(\alpha + \beta)} = \frac{S_P}{\sin\beta} = \frac{S_2P}{\sin\alpha} = \frac{2a}{\sin\alpha + \sin\beta}$

$$\Rightarrow e = \frac{\sin(\alpha + \beta)}{\sin\alpha + \sin\beta} \Rightarrow \frac{e}{1} = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}$$

$$\text{Now } \frac{1-e}{1+e} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$$

$$\therefore \tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{1}{3} \quad \dots(1)$$

Also we know that

$$\cot\frac{\alpha}{2} + \frac{\beta}{2} + \cot\frac{\gamma}{2} = \cot\frac{\alpha}{2}\cot\frac{\beta}{2}\cot\frac{\gamma}{2} = 3\cot\frac{\gamma}{2}$$

$$\Rightarrow 2\cot\frac{\gamma}{2} = \cot\frac{\alpha}{2} + \cot\frac{\beta}{2} \Rightarrow \cot\frac{\alpha}{2}, \cot\frac{\gamma}{2}, \cot\frac{\beta}{2} \text{ are in A.P.}$$

Q43. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is

- (A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

Ans. A

Sol. Equation of chord with mid point (h, k) is $\frac{x}{h} + \frac{y}{k} = 2; m = -\frac{k}{h} \Rightarrow y + mx = 0$.

Q44. The normal to curve $xy = 4$ at the point $(1, 4)$ meets the curve again at :

- (A) $(-4, -1)$ (B) $\left(-8, -\frac{1}{2}\right)$ (C) $\left(-16, -\frac{1}{4}\right)$ (D) $(-1, -4)$

Ans. C

Sol. $xy = 4$

Let normal at $\theta(1, 4)$

$$M_{p\theta} \cdot n \text{ tangent at } p = -1$$

$$xy' + y = 0$$

$$y' = \frac{-y}{x}$$

$$y'_{(1,4)} = -4$$

$$\frac{\left(\frac{2}{t} - 4\right)}{(2t - 1)} \times (-4) = -1$$

$$(2 - 4t)4 = t(2t - 1)$$

$$2t^2 - t = 8 - 16t$$

$$2t^2 + 15t - 8 = 0$$

$$2t(t + 8) - (t + 8) = 0$$

$$(2t - 1)(t + 8) = 0$$

$$t = \frac{1}{2}, \quad t = -8$$

$$\theta\left(-16, \frac{-1}{4}\right)$$

Q45. Locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is a/an

(A) ellipse (B) circle (C) hyperbola (D) parabola

Ans. C

Sol. Let point is (h, k)

C.O.C of circle

$$T = 0$$

$$hx + ky = 4$$

It is tangent to $xy = 1$

$$hx + \frac{k}{x} = 4$$

$$hx^2 - 4x + k = 0$$

$$\text{C.O.T} \quad D = 0$$

$$16 - 4hk = 0$$

$$xy = 4$$

Q46. If the tangent and normal at a point on rectangular hyperbola cut-off intercept a_1, a_2 on x-axis and b_1, b_2 on the y-axis, then $a_1a_2 + b_1b_2$ is equal to

- (A) 2 (B) $\frac{1}{2}$ (C) 0 (D) -1

Ans. C

Sol. Let rectangular hyperbola is $xy = c^2$

$$\left(ct, \frac{c}{t}\right)$$

$$\text{tangent } \frac{x}{x_1} + \frac{y}{y_1} = 2$$

$$\frac{x}{ct} + \frac{yt}{c} = 2$$

$$a_1 = 2ct, \quad b_1 = \frac{2c}{t}$$

equation of normal

$$y - \frac{c}{t} = t^2(x - ct)$$

$$a_2 = ct - \frac{c}{t^3}$$

$$b_2 = \frac{c}{t} - ct^3$$

$$= a_1a_2 + b_1b_2$$

$$= 2ct\left(ct - \frac{c}{t^3}\right) + \frac{2c}{t}\left(\frac{c}{t} - ct^3\right)$$

$$= 2c^2t^2 - \frac{2c^2}{t^2} + \frac{2c^2}{t^2} - 2c^2t^2$$

$$= 0$$